UNIVERSITY OF LONDON

MSci/MSc EXAMINATION 2023

For Internal Students of
Royal Holloway

DO NOT TURN OVER UNTIL TOLD TO BEGIN

CS5100/CS5100J/CS4100: Data Analysis
CS5100R/CS5100JR Data Analysis – for FIRST SIT/RESIT CANDIDATES

Time Allowed: **TWO hours**

Please answer **ALL** questions

CS calculators are permitted

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1. **Principles of Data Analysis (40 marks)**

   (a) **Supervised and Unsupervised Learning.** (10 marks)

   i. What is the difference between supervised and unsupervised learning? [2 marks]

   ii. Is the problem of fitting a straight line to a set of 2-dimensional datapoints, $\mathcal{D} = \{(x_n, y_n) \in \mathbb{R}^d\}_{n=1}^{100}$, a supervised or unsupervised task? Justify your answer. [4 marks]

   iii. Consider the problem of grouping human genes according to the similarities of their DNA sequence. Is this a supervised or unsupervised task? Justify your answer. [4 marks]

   (b) **Training, Testing, and Validation Data Sets.** (10 marks)

   i. Focus on supervised learning problems. Explain why you need to split the available data into training and testing data sets. [4 marks]

   ii. In which cases do you also need a validation data set? Justify your answer and specify if the validation data set should be part of the training or testing set. [3 marks]

   iii. Describe a real-world machine learning task for which you need a validation data set and explain how you use it in practice. [3 marks]

   (c) **Linear Algebra and Principal Component Analysis (PCA).** (10 marks)

   i. Are the vectors $\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 5 \\ -4 \\ -1 \end{bmatrix}$ orthogonal? Explain your answer. [3 marks]

   ii. You are asked to perform PCA on a set of datapoints in $\mathbb{R}^{2023}$. Describe how the scatter matrix of this data set is computed. You should discuss all pre-processing steps in your answer. [7 marks]

   (d) **Density Estimation.** (10 marks)

   i. Expectation-Maximization (EM) algorithm is used to solve the Mixture of Gaussians problem. Is this a parametric density estimation? Explain your answer. [4 marks]

   ii. Describe the curse of dimensionality in the context of density estimation using a histogram. [6 marks]

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2. **KNN (15 marks)**

(a) In 2 lines, explain how the K-Nearest Neighbour algorithm works and why it is called a *transductive* algorithm. [3 marks]

(b) Write the formula of a distance measure, $d : \mathcal{X} \times \mathcal{X} \to \mathbb{R}_+$, that you can use with KNN when the object space is $\mathcal{X} = \mathbb{R}^n$. Make sure your formula shows the dependence on n explicitly. [3 marks]

(c) Describe how you use the training set to predict the label in the regression setup. Specify what happens when $K = 1$. [3 marks]

(d) Describe how you use the training set to predict the label in the classification setup. [3 marks]

(e) Explain the role of the hyperparameter K and how it controls the bias and variance of the algorithm. [3 marks]

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3. **Logistic regression** (15 marks)

(a) In which cases can you use a logistic regression algorithm? Give a real-world example. [3 marks]

(b) The sigmoid function is defined as

$$\sigma(t) = \frac{1}{1 + e^{-t}} \ .$$

What is the domain of the function? And what are the minimum and maximum values that $\sigma(x)$ may assume on that domain? [3 marks]

(c) Show that the derivative of the sigmoid function obeys

$$\sigma'(t) = \frac{d}{dt}\sigma(t) = \sigma(t)(1 - \sigma(t))$$

Hint: Use $\frac{d}{dt}\frac{1}{t} = -\frac{1}{t^2}$, $\frac{d}{dt}e^{-t} = -e^{-t}$, and the chain rule of derivation, i.e. $(f(g(t)))' = f'(g(t))g'(t)$, to compute $\sigma'(t)$. [3 marks]

(d) Explain how you may use the data negative log-likelihood to train a logistic regression model. [3 marks]

(e) Imagine you have the following logistic regression model for predicting a binary random variable $Y \in \{\texttt{green}, \texttt{red}\}$,

$$f(X) = \text{Prob}(Y = \texttt{green}|X) = \sigma(\lambda_0 + \lambda_1 X_1 + \lambda_2 X_2), \quad \lambda = [1, 1, -1]$$

and you are given an unseen test object, $x = [-\frac{1}{3}, 1]^T$. What is the predicted label? Explain how you proceed and show all the steps of your calculation. [3 marks]

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4. **Bagging and Out-of-Bag Error Estimation (15 marks)**

(a) Describe how the ensemble method of bagging is applied to a supervised learning problem, and how the prediction for a test object is made if the problem is a classification problem. [6 marks]

(b) Consider the following data set with 6 datapoints in the form of (x, y):

$$\mathcal{D} = \{(1,4),\ (3,10),\ (4,9),\ (7,10),\ (10,6),\ (12,-1)\}\ .$$

The table shows five bootstrapped training sets (BTS), and the predictions made by the model trained from each BTS.

BTS	Predictions for $x =$					
	1	3	4	7	10	12
$\{(1,4),(1,4),(1,4),(4,9),(7,10),(12,-1)\}$	3	8	11	15	9	0
$\{(3,10),(3,10),(3,10),(4,9),(10,6),(10,6)\}$	5	11	7	7	10	-6
$\{(1,4),(3,10),(7,10),(7,10),(10,6),(10,6)\}$	6	12	7	11	10	-4
$\{(7,10),(10,6),(10,6),(12,-1),(12,-1),(12,-1)\}$	7	9	7	6	5	2
$\{(1,4),(4,9),(4,9),(7,10),(10,6),(12,-1)\}$	2	10	10	8	8	-6

i. What are the out-of-bag (OOB) predictions for $x = 3$, $x = 7$ and $x = 12$? [5 marks]

ii. The OOB predictions for $x = 1$, $x = 4$ and $x = 10$ are 6, 7 and 9 respectively. Use this information and the results in part (i) to calculate the OOB MSE. [4 marks]

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5. **Clustering (15 marks)**

(a) For each of the following statements, determine whether it is true or false, and provide justification. Each justification must have at most 40 words. [9 marks]

i. The K-Means algorithm is a deterministic algorithm.

ii. The K-Means algorithm is used to perform clustering on a data set. If $K = 10$, then the output must contain 10 clusters.

iii. The Agglomerative Hierarchical Clustering (AHC) algorithm is used to perform clustering on a data set. If single linkage is used and we choose 5 output clusters, then it is possible that the furthest pair of datapoints belong to the same cluster.

(b) Let ω_1, ω_2 denote the following two clusters:

$$\omega_1 = \left\{ \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \end{bmatrix} \right\}, \quad \omega_2 = \left\{ \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -4 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \end{bmatrix} \right\}.$$

Compute the complete linkage and the centroid linkage between ω_1 and ω_2. [6 marks]

END

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